

Sulaiman Qazi

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

Sept. 2026 – Apr. 2031

EXPERIENCE

Robotics Software Engineer

Sept. 2026 – Present

WATonomous

Waterloo, ON

- Built a ROS 2 navigation stack in C++ that maps from lidar, plans whole-body A* paths, and steers with pure pursuit
- Added autonomous frontier exploration and cut false map walls from up to 82 per run to 0 by interpolating odometry

Machine Learning Research Intern

July 2026 – Aug. 2026

Sunnybrook Research Institute - Focused Ultrasound Lab

Toronto, ON

- Built a PyTorch U-Net (2,500 lines) that finds red blood cells in tissue slides, hitting a >0.79 Dice score against hand labels on unseen data
- Validated the pipeline on synthetic cells first, hitting a 0.99 Dice score to rule out code bugs before real data
- Ran roughly 30 controlled experiments and showed only 2 of 11 techniques tested made a real difference
- Used AI coding agents (Claude Code) to script, launch, and monitor experiments, including overnight training runs

Hardware/Firmware Engineering Intern

July 2025 – Aug. 2025

Sunnybrook Research Institute - Focused Ultrasound Lab

Toronto, ON

- Wrote embedded C++ firmware for an RP2040 replacing a manual switch-and-knob test rig with scripted control of 8 MOSFET-switched channels and a quad digital potentiometer over I2C
- Built a computer interface for the board so failure scenarios could be run automatically, with no operator at the bench
- Added a hardware emergency-stop and a trigger that syncs the rig with other equipment, with a status light
- Reverse-engineered the undocumented predecessor board, then authored the 15-page hardware design specification
- Designed the 4-layer replacement PCB in Altium with dedicated power and ground pours, cutting cost 33%
- Hand-soldered and bring-up tested 200+ SMD and through-hole components against firmware behaviour

PROJECTS

Game Boy Emulator | C, SDL2, CMake

- Emulated the Sharp LR35902 CPU, decoding each ROM byte into its opcode and executing the full instruction set
- Built the hardware as separate modules (CPU, memory/MMU, I/O, and PPU) communicating over a shared bus
- Rendered graphics through SDL2, running commercial ROMs (Tetris, Mortal Kombat 3) at playable frame rates
- Built a Chip-8 interpreter first to learn the fetch-decode-execute loop before tackling the Game Boy

Health Tracking Smartwatch | ESP32, Altium, C++

- Designed a custom PCB integrating an ESP32, sensor array, and TFT display, reflow-soldered by hand
- Programmed heart-rate and step-tracking firmware with sensor sampling and a TFT_eSPI display UI

LEADERSHIP & AWARDS

Programmer

Feb. 2025 – June 2026

FTC 22603 Hurricane Stormbots

Ajax, ON

- Programmed autonomous odometry navigation in Java; team won 1st place Inspire Award for overall achievement

Co-President - Math, STEM, and CS Clubs

Sept. 2024 – June 2026

J. Clarke Richardson Collegiate

Ajax, ON

- Ran weekly meetings, activity planning, and attendance across three clubs totalling 250+ members

Hackathons: 1st place, Hack the Valley Hack Day | Ribbon Podium, Hack the 6ix

Competitions: 2nd of 11 teams, Durham Coding Contest | Think Award, FTC Canada Cup

TECHNICAL SKILLS

Languages: C++, C, Python, Java, JavaScript, MATLAB, SQL, PHP, HTML/CSS

Frameworks & Tools: PyTorch, NumPy, ROS 2, Docker, Google Colab, SDL2, Git, GitLab, Arduino, Linux

Hardware: PCB Schematic & Layout (Altium, KiCad), I2C, SPI, UART, SMD Soldering, SOLIDWORKS